CHITRANG PATEL

Software Engineer | C# / .NET | Backend & Real-Time Systems
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PROFESSIONAL SUMMARY

Software Engineer with 8+ years delivering production C# applications, real-time client-server systems, and interactive platforms shipped across iOS and Android. Strong in object-oriented programming, SOLID principles, and software architecture, with hands-on REST API development and integration, WebSocket-based real-time communication, database management, and performance optimization. Cut development cycle time ~30% through reusable, modular design and disciplined code review. Daily practitioner of AI-assisted development, applying LLM tooling to code generation, debugging, testing, and documentation. Collaborative across design, QA, and product in Agile environments.

CORE COMPETENCIES

Software Engineering • Backend Development • Full Stack Development • API Development & Integration • Real-Time Distributed Systems • OOP • Software Architecture • System Design • Database Design • Performance Optimisation • Debugging • Unit Testing • Code Review • Agile / Scrum • SDLC • CI/CD • Technical Documentation • Mentoring

TECHNICAL SKILLS

Languages: C#, SQL, JavaScript, TypeScript
Backend & APIs: .NET, REST API design and integration, WebSocket / real-time messaging, Node.js, Express.js, client-server architecture, authentication flows, third-party SDK integration
Databases: Microsoft SQL Server, SQL, Firebase (NoSQL), data modelling, query optimisation
Architecture & Practices: OOP, SOLID, design patterns, modular and component-based architecture, state machines, unit testing, performance profiling, memory optimisation
Tooling & Platforms: Git, CI/CD pipelines, Agile/Scrum, code review, Visual Studio, Unity Engine (C#), iOS, Android, AR/VR, cloud-backed services

AI & DEVELOPER PRODUCTIVITY SKILLS

- **AI-Assisted Development & Prompt Engineering:** Daily use of LLM tooling (ChatGPT, Claude, GitHub Copilot, Cursor) for code generation, refactoring, and design exploration; structuring prompts and context for production-usable output
- **AI-Assisted Debugging & Testing:** Faster root-cause analysis on unfamiliar codebases; generating unit test scaffolding, edge cases, and regression checks
- **AI-Assisted Documentation & Workflow Automation:** API docs, technical specs, and onboarding guides; automating repetitive engineering tasks and boilerplate generation
- **Generative AI Literacy:** Practical grasp of LLM capabilities and limits, including Model Context Protocol (MCP) for tool-connected AI workflows

PROFESSIONAL EXPERIENCE

Software Engineer — ViitorCloud Technologies, Ahmedabad | *April 2023 – Present*

- Cut feature development time **~30%** by designing reusable, extensible modules governed by OOP and SOLID principles, enabling consistent on-time delivery against every client commitment
- Architected modular, component-based systems that lowered regression risk and shortened the path to shipping new features on existing products
- Engineered real-time client-server features over authoritative networking and socket-based messaging, handling state synchronisation, latency tolerance, and connection recovery
- Integrated REST APIs and third-party SDKs with structured error handling, retry logic, and payload validation for reliable production behaviour
- Optimised runtime performance and memory footprint via profiling, asset streaming, and object pooling, improving load times and stability on constrained hardware
- Embedded AI-assisted development across coding, debugging, and documentation, measurably reducing cycle time; recognised as **Entrant of the Quarter**

Software Engineer — nCore Games, Bangalore | *December 2020 – March 2023*

- Owned core application logic, decision-making AI, statistics, and mathematics subsystems for a globally released, officially licensed sports simulation product
- Designed an AI decision engine using goal-oriented action planning, translating probabilistic models into deterministic, testable, tunable behaviour
- Built the statistics and mathematics layer, guaranteeing numerical correctness across a large simulation state space
- Reviewed peer code through CI/CD-gated pull requests, raising baseline quality and reducing defects reaching QA; delivered Agile sprint commitments ahead of schedule

Senior Unity Game Developer — Vasundhara Vision, Surat | *December 2019 – December 2020*

- Delivered **5+ production applications** for iOS and Android from architecture through store release, owning technical decisions end to end
- Built online multiplayer backend services with Node.js, Express.js, and Socket.IO — session and room management, authoritative server logic, real-time event broadcasting
- Implemented in-app purchases, leaderboards, and ad network integrations with server-side transaction validation
- Led a small engineering team: set code standards, ran reviews, mentored developers, and streamlined workflows to meet every release date

Unity Game Developer — Escrow Infotech, Surat | *October 2017 – October 2019*

- Designed and implemented application logic, interactive UI, and animation systems for consumer mobile products on Android
- Drove a **20% increase in user retention** through iterative UX refinement informed by user feedback and store review analysis
- Established reusable UI and animation components adopted across multiple projects, eliminating duplicated implementation effort

KEY PROJECTS

SoReal — Interactive Real Estate Visualisation Platform | *C#, .NET, real-time networking, REST APIs*

- Owned the technical spec, interaction systems, multi-user session handling, and full client application layer for a 3D property-walkthrough platform used in live client engagements
- Solved multi-user state synchronisation over unreliable connections while holding large 3D scenes inside mobile memory budgets

ICC Cricket Mobile — Licensed Sports Simulation | *C#, OOP, algorithms, state machines, CI/CD*

- Built batting, bowling, and fielding mechanics and authored the AI decision layer from scratch using goal-oriented action planning, plus statistics and mathematics subsystems
- Balanced statistically believable AI against the determinism needed for unit testing and tuning; core modules delivered ahead of schedule for a global release

Ludo Multiplayer — Real-Time Real-Money Gaming Platform | *C#, Node.js, Express.js, Socket.IO, REST APIs*

- Implemented the full game state machine, API integration layer, socket event handling, bot decision logic, and transaction-facing error management across online and offline modes
- Guaranteed transactional correctness and fair play under concurrent real-money sessions with graceful mid-session disconnect recovery

Configurable Card Game Engine | *C#, data-driven configuration, IAP, ad SDK*

- Designed a data-driven rules engine supporting eight rule variants toggleable at runtime, with AI bot opponents, animation pipeline, and monetisation integrations
- Abstracted rules flexibly enough for arbitrary combinations without conditional-logic sprawl; two titles shipped from one reusable engine, cutting build time on the second substantially

Additional Shipped Titles — Shooting Arena: Total Destruction, 99.9% Impossible Bus Simulator | *C#, physics simulation, state-driven AI, IAP/ads*

- Built player control systems, enemy AI, health/damage systems, and a virtual currency economy resistant to exploitation; implemented vehicle physics and a content pipeline for rapid level authoring, tuned across a wide range of device performance profiles

EDUCATION

Bachelor of Engineering, Computer Engineering — Dr. D. Y. Patil College of Engineering, Pune | *July 2012 – May 2016*

Relevant coursework: Object-Oriented Programming, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks

ACHIEVEMENTS & LANGUAGES

- **Entrant of the Quarter**, ViitorCloud Technologies — recognised for engineering contribution and consistent delivery performance
- Shipped **10+ production applications** across iOS and Android over 8+ years, spanning solo ownership through team leadership; reduced team development time **~30%** via reusable architecture patterns adopted across projects
- **Languages:** English (Professional) | Hindi (Native) | Gujarati (Native)